



US012414880B2

(12) **United States Patent**
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(10) **Patent No.: US 12,414,880 B2**

(45) **Date of Patent: Sep. 16, 2025**

(54) **TAMPON WITH WICKING MEMBER
ADAPTED FOR IMPROVED
MANUFACTURABILITY AND WICKING
PERFORMANCE**

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 1156 days.

(21) Appl. No.: **17/191,887**

(22) Filed: **Mar. 4, 2021**

(65) **Prior Publication Data**

US 2022/0280347 A1 Sep. 8, 2022

(51) **Int. Cl.**
A61F 13/20 (2006.01)
A61F 13/15 (2006.01)

(52) **U.S. Cl.**
CPC **A61F 13/2054** (2013.01); **A61F 13/208**
(2013.01); **A61F 2013/15959** (2013.01)

(58) **Field of Classification Search**
CPC A61F 13/2054; A61F 13/208; A61F
2013/15959
See application file for complete search history.

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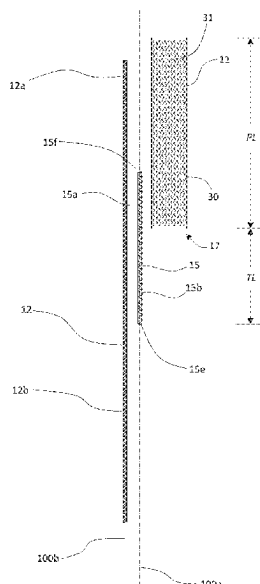
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(57) **ABSTRACT**

A tampon including a pledget, a wicking member and a withdrawal cord is disclosed. The pledget includes a mass of fibrous absorbent material disposed within or between one or more layers of an outer cover formed of a first nonwoven web material. The wicking member includes a second nonwoven web material distinct from the first nonwoven web material and the withdrawal cord. The wicking member is joined in direct contact with a surface of the pledget, and has a width less than the width of the pledget width.

20 Claims, 9 Drawing Sheets



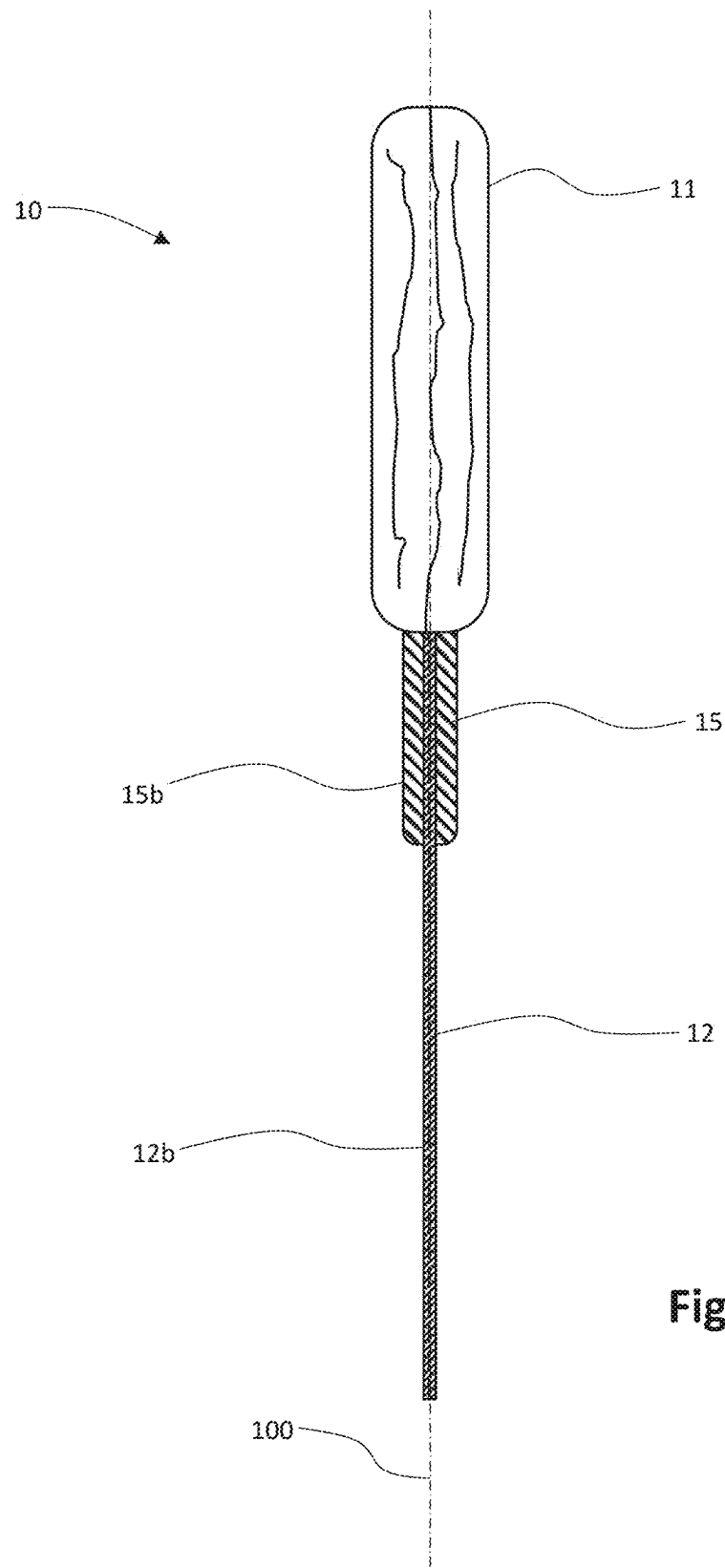


Fig. 1

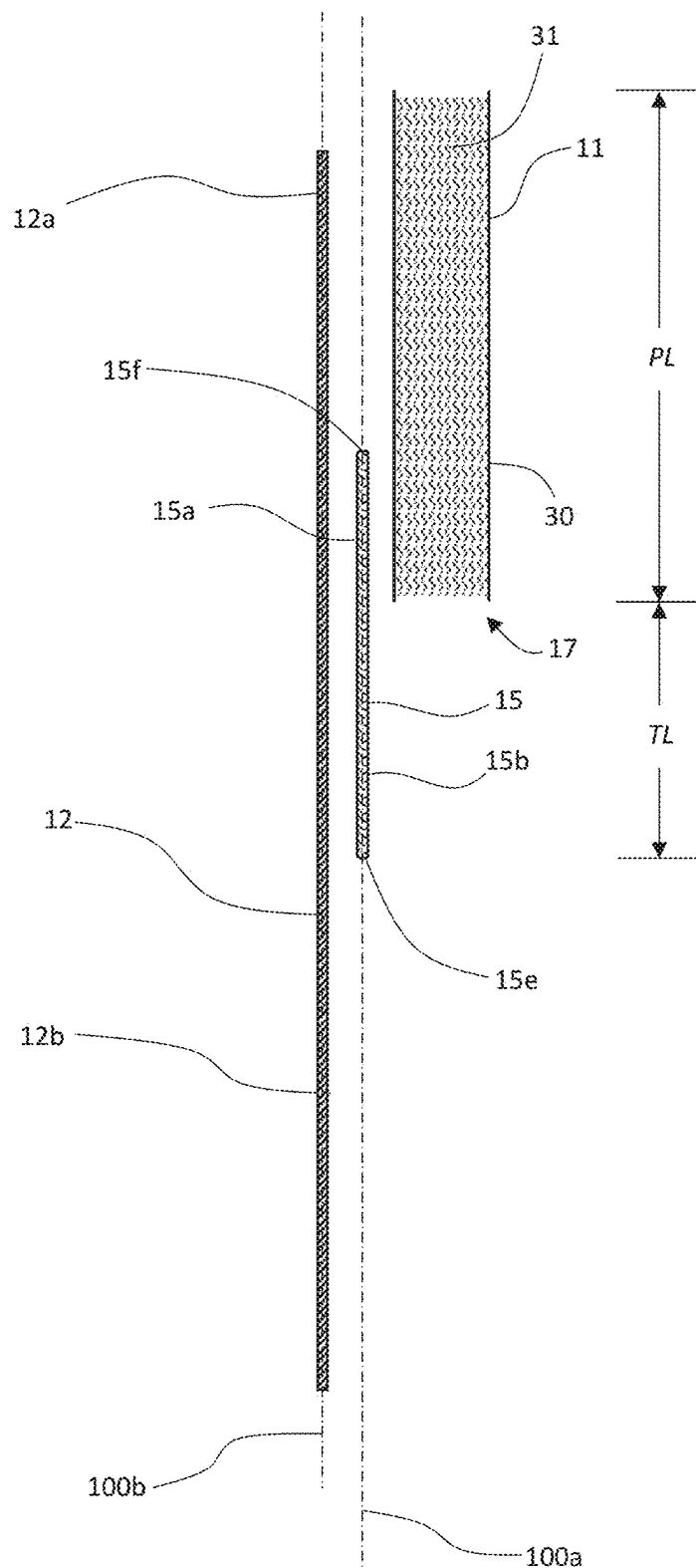


Fig. 3B

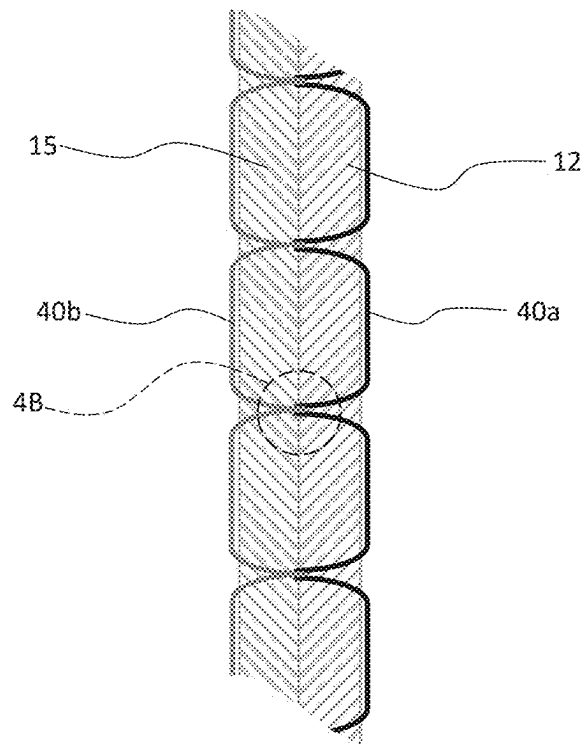


Fig. 4A

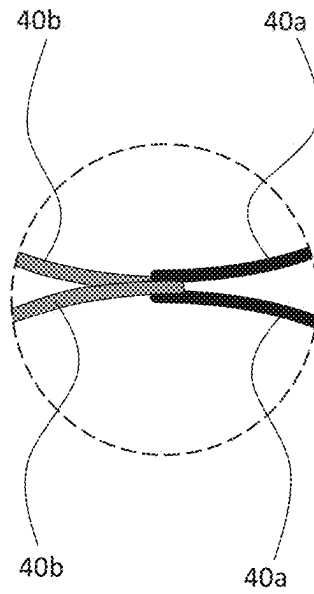


Fig. 4B



Fig. 5A

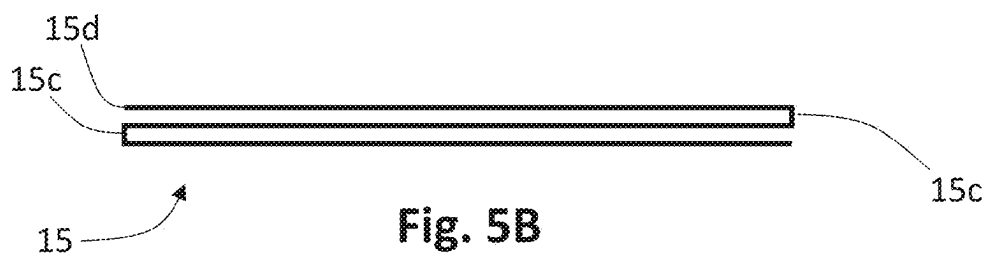


Fig. 5B

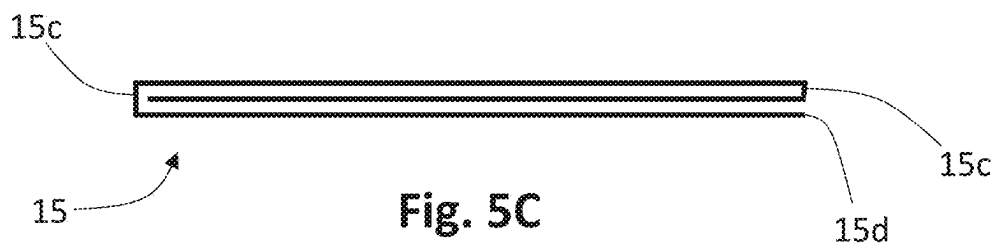


Fig. 5C

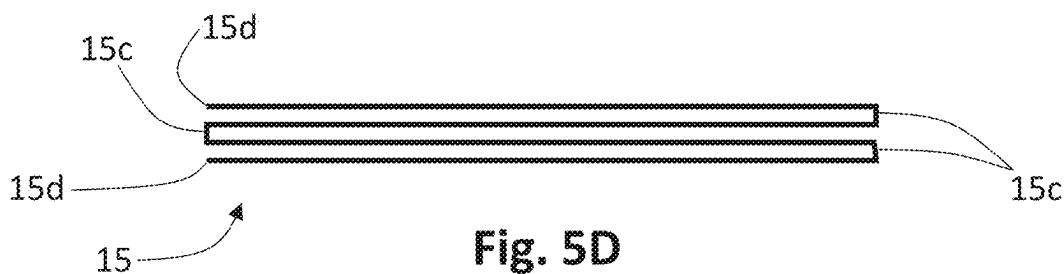


Fig. 5D



Fig. 5E

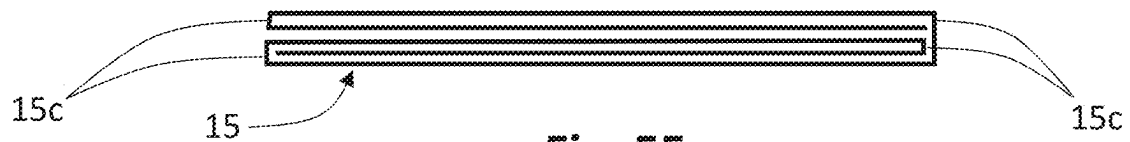


Fig. 5F

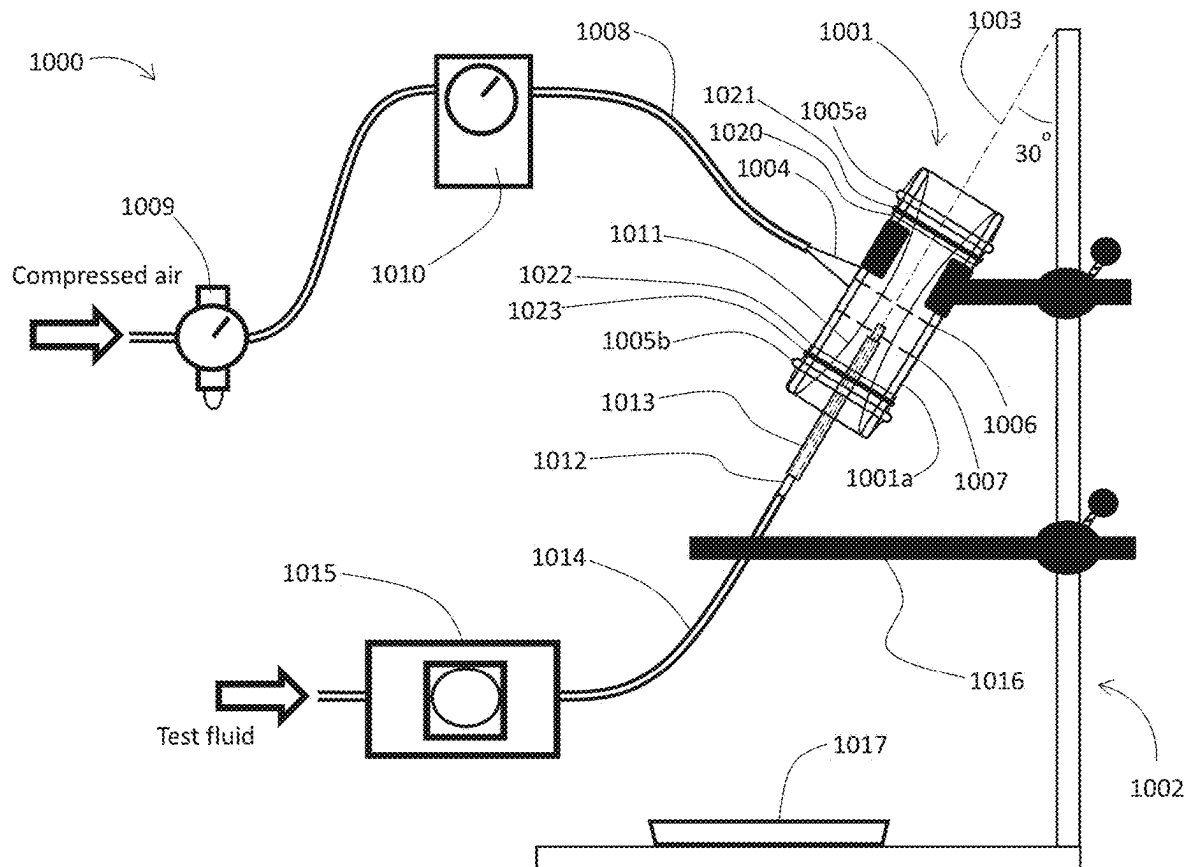


Fig. 6A

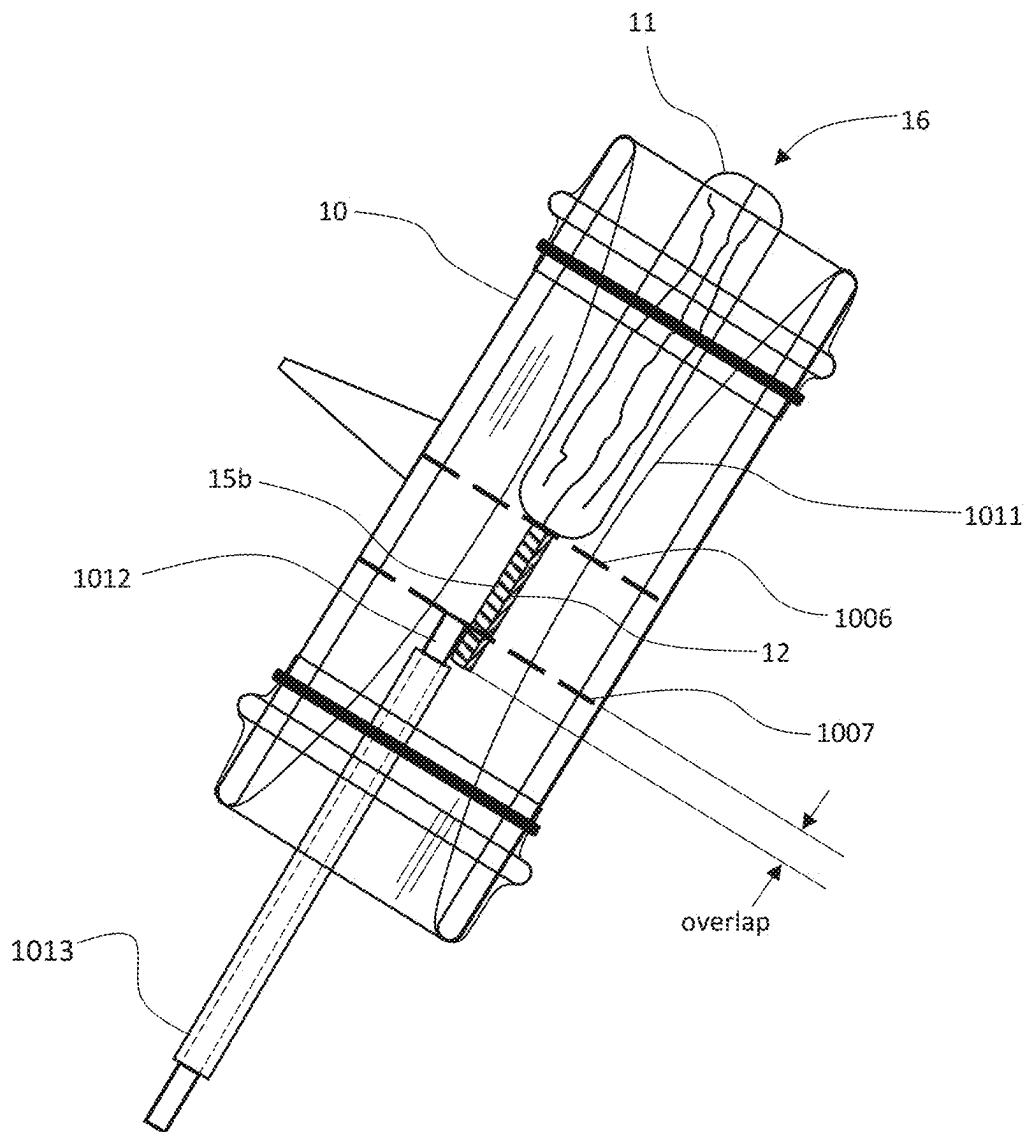


Fig. 6B

1

TAMPON WITH WICKING MEMBER ADAPTED FOR IMPROVED MANUFACTURABILITY AND WICKING PERFORMANCE

BACKGROUND OF THE INVENTION

A variety of designs for absorbent tampons have been manufactured for a number of years, and used by women to capture and absorb menstrual fluid internally, in conjunction with, or as an alternative to, externally worn feminine hygiene pads. Many women prefer to use tampons as an alternative to feminine hygiene pads at least some of the time during menstruation. Among other reasons for such preference, because tampon use is internal it is more discrete, avoiding the bulk under clothing that is associated with many types of feminine hygiene pads.

Particularly when a feminine hygiene pad is not used in conjunction therewith, it is important that the tampon capture and absorb most if not substantially all of the menstrual fluid that is discharged during the tampon's usage duration (to the extent of its absorption capacity), to help avoid a leakage of fluid that may soil underwear, outer clothing, bedclothes, etc. The prior art has recognized various ways in which tampons might fail to perform effectively. One such way is sometimes referred to as "bypass" failure. Bypass failure occurs when the menstrual fluid travels along the walls of the vaginal cavity without contacting the tampon, or the tampon, while having available absorption capacity, otherwise fails to capture and absorb the fluid.

A variety of approaches to tampon design have sought to mitigate such failure. One approach that has proven effective has been to include a wicking member as part of the withdrawal cord. The wicking member is a material/structure selected and configured to extend downward (or trail, to the rear of) the main pledget, along with the withdrawal cord, thereby extending further down the vaginal cavity toward the vaginal opening than the pledget, following insertion. Appropriately configured, the wicking member can engage menstrual fluid flowing along the vaginal cavity past the pledget, capture it, and wick it back to the pledget. Current approaches to manufacture of such tampons, however, have shown to be inefficient, and current selections of configurations and materials have shown to be less effective than might be hoped.

Accordingly, there remains room for improvement in the construction and method of manufacture of tampons with wicking members.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a longitudinal side view of an example of a tampon with a pledget shaped and compressed into a self-sustaining form.

FIG. 2 is a longitudinal front side view of an example of a tampon before the pledget has been shaped and compressed into a self-sustaining form.

FIG. 3A is a lateral cross section of the tampon shown in FIG. 2, taken through line 3A-3A in FIG. 2.

FIG. 3B is an exploded longitudinal cross section of components of the tampon shown in FIG. 2, taken through line 3B-3B in FIG. 2, and shown with the components separated, i.e., not joined together by stitching or other joining mechanism.

FIG. 4A is a longitudinal cross section of portions of a wicking member and withdrawal cord of the tampon shown in FIG. 2, taken through line 4A-4A in FIG. 2.

2

FIG. 4B is an enlarged view of portion of lockstitching shown within circle 4B in FIG. 4A.

FIGS. 5A-5F agree schematic lateral cross sectional views of folding configurations for a wicking member formed of a web material.

FIGS. 6A and 6B are schematic depictions of a configuration of equipment used in the Wicking Measurement method herein.

DETAILED DESCRIPTION OF EXAMPLES

Definitions

As used herein the term "tampon" refers to any type of absorbent structure which is inserted into the vaginal cavity for the absorption of fluid therefrom. Typically, a tampon includes a pledget structure including a quantity of absorbent material, often absorbent fibrous material, which pledget structure has been bunched, folded and/or compressed in one or more lateral/radial directions, the longitudinal direction, or both, via application of pressure, heat and moisture control, in order to provide a formed tampon having a size, shape (typically cylindrical) and stability of form to facilitate insertion into the vagina. A tampon which has been so formed is referred to herein as a "self-sustaining" form. The degree of compression, heat and moisture control applied to the pledget is sufficient such that in the subsequent absence of the external forces and absence of substantial contact with moisture, the pledget will tend to retain its general formed shape and size. It will be understood by persons of ordinary skill in the art that this self-sustaining form typically does not persist following insertion of the tampon. Once the tampon is inserted and begins to contact and absorb fluid, the pledget will swell with absorbed fluid, expand and lose its self-sustaining form.

As used herein the terms "pledget" or "tampon pledget" are intended to be interchangeable and refer to a structure including absorbent material configured to perform the primary function of the tampon, absorption of menstrual fluid. A tampon pledget is sometimes referred to as a tampon blank, or a softwind, and the term "pledget" is intended to include structures designated by such terms as well.

As used herein "vaginal cavity" refers to the internal space within the genitalia of the human female, located between the introitus of the vagina (sometimes referred to as the sphincter of the vagina) and the cervix.

With respect to fiber components of a nonwoven web material, "biased" or "directionally biased" along a longitudinal direction means that a greater proportion of the lengths of the fiber components, in the aggregate, are more aligned with the longitudinal direction than with the lateral direction. As a result of methods of manufacture of spunbond and carded nonwoven materials, for example, their fiber components tend to be accumulated and consolidated in a manner that imparts them with a directional bias along the direction of manufacture (i.e., the machine direction). Thus, for example, if a spunbond nonwoven is used as a component of a tampon, the fibers will be directionally biased along the longitudinal direction if the machine direction of the nonwoven is oriented along the longitudinal direction of the tampon.

With respect to a tampon, the "longitudinal" direction is the ordinary general direction of ejection from an applicator; and also corresponds with the ordinary general direction of insertion of a tampon into and its withdrawal from the vaginal cavity in normal use. For a completely manufactured, pre-use tampon that has a pledget with a generally

cylindrical or capsule-shaped self-sustaining form, the longitudinal axis of the form generally lies along the longitudinal direction. A “radial” or “lateral” direction is a direction perpendicular to the longitudinal direction. References to “length” herein refer to a dimension along the longitudinal direction; references to “width” herein refer to a dimension along the lateral direction.

A “nonwoven,” “nonwoven web,” “nonwoven web material,” or “nonwoven fabric” is a cloth-like web material (or portion or section thereof) formed predominantly of fibers that are neither knitted nor woven, but rather, laid down and accumulated to a desired basis weight, then consolidated and held together to form a web, via one or any combination of calendering, thermal and/or compression bonding, bonding via use of a binder, heating (via, e.g., heated air driven through an accumulation of fibers) or hydroentangling (spunlace). The predominant fibers may be natural fibers harvested from plant material (e.g., cotton) (but excluding tree wood pulp), semi-synthetic (e.g., rayon, lyocell, viscose), or synthetic (e.g. fibers spun from molten polymer resin(s)), or any combination thereof. Herein, a skin- or membrane-like film (e.g., extruded or otherwise formed from polymer resin(s)) is not deemed a nonwoven. Herein, a paper tissue product, paper product, or paperboard or cardboard product, formed via wetlaying and predominately constituted of tree wood pulp, is not deemed a nonwoven.

“Opened configuration,” with respect to a tampon, means the configuration of the tampon prior to the time it is compressed and formed into a self-sustaining form during manufacture, or in the case of a finished product, after it is completely ejected from an applicator (if present) and/or allowed and/or caused by any suitable technique to open and substantially re-assume its pre-compression shape and size.

“Predominant,” and forms thereof, when used to characterize a quantity of a component present in a material, means that a majority of the weight of the material is constituted by the component.

“Withdrawal cord” refers to any section of string, yarn, cord, ribbon, strip material or other flexible/pliable elongate structure typically (although not necessarily) formed of fibrous material, attached to and/or extending from a tampon pledget and trailing from its rearward end. A withdrawal cord of sufficient length may be provided with a tampon for the purpose of providing a relatively thin and flexible trailing member of sufficient length to allow for a portion thereof to trail and remain outside of the introitus following full insertion of the tampon, which the user may easily grasp and pull to withdraw the tampon from her body following a desired duration of use.

The present disclosure relates to an improved absorbent tampon provided with a leakage protection feature. It has been found that there are several potential mechanisms beyond simple bypass flow which may contribute to tampon leakage. Without wishing to be bound by theory, some of these mechanisms may be explained by the following observations. It has been found that many current tampons show stains along the length of the withdrawal cord following use and withdrawal, associated with incidents of tampon leakage. It is believed that the withdrawal cord of many current tampons may offer an escape route for menstrual fluid present at the base of the vaginal cavity, by operating as a wicking mechanism.

During a tampon change, some residual menstrual fluid may be left in the vaginal cavity near the introitus. This may be fluid that was absorbed by the tampon being removed, but is subsequently expressed from the tampon as it is drawn out of the body through the relatively narrow sphincter of the

vagina. Such residual fluid, particularly if located near the introitus (i.e. in the lower vaginal cavity) may not be effectively absorbed by the replacement tampon. This is particularly true of many current tampons which are typically inserted somewhat more deeply into the vaginal cavity. These circumstances, as well as bypass leakage described above, and other leakage circumstances are addressed by tampons within contemplation of the present disclosure.

Pledget and Withdrawal Cord

FIGS. 1, 2, 3A and 3B show one non-limiting example of such an absorbent tampon 10, having a longitudinal axis 100, a pledget 11 having a forward end 16 and a rearward end 17 and a withdrawal cord 12 attached to the pledget and having a leading portion 12a attached to the pledget and a trailing portion 12b extending rearward from a location proximate the rearward end 17. Tampons contemplated herein, however, are not limited to structures having the particular configuration shown in the drawings.

The pledget 11 of the tampon 10 as shown in FIGS. 1 and 2 has a forward end 16 and a rearward end 17. During manufacture of the tampons the pledget 11 may be folded, bunched, compressed and/or otherwise formed in size and shape, from its initially manufactured configuration (e.g. as shown in FIGS. 2 and 3A) into a generally cylindrical and/or capsule-shaped configuration (e.g. as shown in FIG. 1) in the radial direction, the longitudinal direction, or in both the radial and longitudinal directions. While the pledget 11 may be formed into a substantially cylindrical and/or capsule-shaped configuration as suggested in FIG. 1, other shapes are also possible. These may include shapes having a lateral cross section which may be described as oval, elliptical, ovoid, stadium, rectangular, triangular, trapezoidal, semi-circular, or other suitable shapes.

The pledget contemplated herein may have any suitable form and structure, for example, as depicted in FIGS. 2, 3A and 3B. Other non-limiting examples of suitable pledget forms, material composition and structure are depicted and described in US2010/0268182 and US 2007/0260211.

A wicking member 15, described in greater detail below, may be joined to either a withdrawal cord 12, the pledget 11 itself, or both. This joining of the wicking member may occur subsequently to compression of the pledget 11 to a self-sustaining form. In some variations it may be desirable to attach some or all of the wicking member 15 to the pledget 11, the withdrawal cord 12, or both, prior to compression of the pledget 11 to a self-sustaining form. In one method of making of a tampon 10, described more fully below, the wicking member 15 may be integral with the pledget 11 prior to compression of the pledget. In any of the above mentioned manners of construction, the trailing portion 15b of wicking member 15 is preferably not compressed with the pledget 11; or, if compressed, is not compressed to the same degree as the pledget 11.

Prior to formation into a self-sustaining form, the pledget 11 may be of any suitable shape, size, material, or configuration. In the non-limiting example shown in FIGS. 2, 3A and 3B, pledget 11 includes a batt or other mass of absorbent material 31, disposed within an outer wrapper 30. This type pledget may be formed on a continuous processing line wherein absorbent fibrous material is continuously deposited (e.g., via an airlaying process) to form a continuous batt having a desired cross-direction width and depth/weight, on a continuous web of wrapper material being conveyed along a machine direction. The wrapper material web may then be wrapped about the batt by suitable web guiding equipment, and affixed to itself via, e.g., adhesive, to form a continuous wrapped batt. Individual pledgets may then be cut from the

5

continuous batt by repetitive die cutting across the moving batt (i.e., cutting along the cross direction). The cross-direction cuts may be linear, which will result in rectangular pledgets. Alternatively, as suggested in FIG. 2, the cross-direction cuts may be non-linear; in the example depicted, the cutting tool may be configured to make cuts forming the respective forward and rearward ends of each successive pledget, having an arched or curved profile. For a pledget of the configuration depicted, this curved profile, or in a slight modification, a cut profile that will impart the uncompressed pledget with a chevron shape, helps facilitate subsequent compression and formation into a cylindrical or capsule-shaped form with rounded or otherwise tapered forward and rearward ends, through the graduating reduction or tapering down, via the cut profile, in the bulk/quantity of material that must be compressed at each end. Other shapes that embody a tapering down of the quantity of material present toward the forward and rearward ends of the pledget are also contemplated.

While the pledget 11 shown in FIG. 2 is approximately chevron-shaped, other shapes such as but not limited to rectangular, trapezoidal, triangular and hemispherical may be used for tampons within contemplation of the present disclosure. It may be desired, however, that the cut profile be configured to form respective rear and front cut ends of respectively leading and trailing pledgets being cut from the batt as it moves through the manufacturing line, with no generation of cutoff waste/scrap. It can be appreciated that the non-limiting example of an end cut profile reflected in FIG. 2 provides this benefit. Because such a shape also embodies a tapering down of the quantity of material present in lateral cross-section approaching the forward and rearward ends, such shape also facilitates compression into self-sustaining form that is cylindrical, with rounded forward and rearward ends.

In other examples (not specifically shown), the pledget 11 may be a laminar structure including integral or discrete layers. As noted, in the example shown in FIGS. 2, 3A and 3B, the pledget 11 may include an enveloping wrapper 30 and one or more layers of absorbent material 31 positioned within the wrapper. In other examples, the pledget need not have a layered structure at all. To facilitate compression into its self-sustaining form the pledget 11 may be folded, e.g., as depicted herein, may be rolled (e.g., as in currently marketed U BY KOTEX brand tampons, a product of Kimberly-Clark Worldwide, Inc., Irving, TX), may comprise a "petal" structure (e.g. of overlaying/underlaying, crossing rectangular patches of absorbent material, in a configuration present in PLAYTEX SPORT brand tampons, a product of Edgewell Personal Care LLC, Chesterfield, MO) or any other of the structures and configurations which are known in the art relating to tampon pledgets and their manufacture.

The pledget 11 and absorbent material 31 therein may include a wide variety of liquid-absorbing materials commonly used for absorbency in absorbent articles, such as rayon fiber, cotton fiber, or comminuted wood pulp fiber (sometimes called "airfelt"). Examples of other suitable absorbent materials may include creped cellulose wadding; spun and/or meltblown polymer fibers or filaments; chemically stiffened, modified or cross-linked cellulosic fibers; other synthetic fibers such as polyamide fibers (e.g., nylon fibers); peat moss; absorbent foams (such as open-celled foam formed through polymerization of a high internal phase water-in-oil emulsion); nonwoven web materials of natural and/or synthetic fibers or combinations thereof, tissue including tissue wraps and tissue laminates; or any equivalent material or combinations of materials, or blends

6

or combinations of these. Suitable rayon fibers may include but are not limited to viscose, MODAL, TENCEL (or lyocell); tri-lobal and conventional rayon fibers, and needle punched rayon). Suitable cotton fibers may include long fiber cotton, short fiber cotton, cotton linters, T-fiber cotton, card strips, and comber cotton. Preferably, the cotton fibers or fabric layer thereof should be scoured (for removal of natural hydrophobic waxes and impurities) and bleached (for whiteness) and may be imparted with a glycerin finish (for enhancing compaction), a leomin finish (for lubricity), or other suitable finish. Additionally, superabsorbent materials, such as superabsorbent polymers or absorbent gelling materials may be incorporated into the pledget. In particular examples it may be desired that rayon or cotton or a blend thereof, form the greater proportion (by weight) of the absorbent material 31, or that rayon alone form the greater proportion (by weight) of the absorbent material 31, since rayon fibers may possess absorbency properties or capacity greater than those of other fibrous materials, per unit weight and/or per unit cost.

In the example shown in FIGS. 2, 3A and 3B, the pledget 11 may be formed of a body of soft absorbent fibrous material such as rayon fibers or cotton fibers or a combination or blend thereof, and the wrapper 30 may be formed of a woven, knitted or nonwoven web fabric material of suitable composition. The materials for the body may have the form of nonwoven or woven fabric or a batt formed by any suitable process such as airlaying, carding, wetlaying, hydroentangling, or other known fiber deposition and consolidation techniques.

The absorbent material of the pledget 11 may be surrounded with a liquid permeable wrapper 30. Wrapper materials may include rayon, cotton, spunbond monocomponent, bicomponent or multicomponent fibers spun from polymer resins, or other suitable natural or synthetic fibers known in the art. If the pledget 11 is layered, the layers may include different materials. For example, in the example shown in FIG. 2 the wrapper 30, may be constituted primarily of rayon, while the absorbent material 31 may be constituted primarily of cotton. In other examples the wrapper may be constituted primarily of cotton, and the intermediate layer or layers may be constituted primarily of rayon. Optionally, the entire pledget 11 may be formed of a uniform or nonuniform blend of materials throughout. In another particular example, wrapper 30 may be formed of a nonwoven web of spunbond fibers. The spunbond fibers may be spun from, for example, polymer resin including polyolefins such as polypropylene, polyethylene, or a blend or combination thereof. In a more particular embodiment the spunbond fibers may be spun bicomponent fibers including a first polypropylene resin component and a second differing polypropylene resin component or a polyethylene resin component. When formed of ordinarily hydrophobic materials such as polyolefins (including polypropylene and polyethylene) wrapper 30 material may be treated, e.g., by application of a suitable surfactant, to render it hydrophilic, so that it will readily attract and permit aqueous fluid to wick therethrough to the absorbent material within the wrapper. A nonwoven web material formed of polymeric material as described may be desired to form the wrapper, over natural fibrous materials or semi-synthetic rayon, for reasons of having a soft, smooth and comfortable feel and low friction against sensitive skin and internal tissues, relatively low cost and superior wet structural integrity.

The pledget 11 may have any suitable size, shape and thickness that will both provide a suitable quantity of absorbent material and resulting absorption capacity, while

permitting compression into a self-sustaining form of a size and shape suitable for easy and comfortable insertion. An uncompressed, opened size similar to those of conventional currently available tampons has been found to work well. A typical size for an uncompressed pledget may be from about 2 cm to about 8 cm in longitudinal length and from about 3 cm to about 8 cm in lateral width, including any combination of length and width within those ranges, in combination with an uncompressed thickness anywhere from about 1 cm to about 3 cm. Total basis weight for a flat, uncompressed and open pledget, may be from about 150 g/m² to about 1,400 g/m², calculated as the weight of the pledget divided by the largest surface area on one side of the pledget. Optionally, a pledget 11 that is shorter and wider than the ranges given above may also be desired in some circumstances to promote relatively greater swelling/expansion in a lateral or radial direction during use.

A withdrawal cord 12, configurations of which are depicted in the figures, is preferably joined to the pledget to facilitate withdrawal of the tampon from the vagina following a desired duration of use. The withdrawal cord 12 may have a leading portion 12a joined to the pledget 11 and a trailing portion 12b extending beyond the rearward end 17 thereof. In other examples, the withdrawal cord may be integral with the pledget, or an extension of a structural component of the pledget, such as of an overwrap as described above. In some examples the withdrawal cord 12 may be integral with a wicking member 15.

In a particular example, the withdrawal cord 12 may be a separate section of cord, string, yarn, ribbon, knitted cord or strip of woven or nonwoven fabric formed separately of the components of the pledget and wicking member, and then attached by any suitable mechanism to the pledget and/or to the wicking member.

The attachment mechanism may include sewing, adhesive attachment, thermal or pressure bonding, through-pledget punching, penetration and/or looping of the withdrawal cord material about structure(s) of the pledget or portions thereof, or any combination of these. A leading portion 12a of the withdrawal cord 12 may be attached or joined to any suitable location on the pledget 11, although it may be preferable that the attachment/joining location be substantially laterally centered on the pledget and proximate to, or include a location proximate to, the rearward end 17 of the pledget, so that tensile withdrawal force in the cord, exerted by the user, acts predominately on the rearward end of the pledget and does not tend to substantially rotate or reorient the pledget within the user's body during withdrawal. In the example shown in FIGS. 2, 3A and 3B, a leading portion 12a of the withdrawal cord 12 is joined to the pledget 11 along the length of the pledget 11, and trailing portion 12b trails free beyond the rearward end 17 of the pledget 11. The withdrawal cord 12 may be attached to the tampon pledget 11 while the pledget 11 is still uncompressed, as shown in FIG. 2. The withdrawal cord 12 may be attached along substantially the entire length of one major surface of the pledget 11.

To minimize chances of failure of the attachment between the withdrawal cord 12 and the pledget (i.e., separation) during withdrawal, it may be desired that the withdrawal cord be directly or indirectly attached along substantially the entire length of the pledget, thereby diffusing tensile withdrawal force exerted by the user, by distributing it over the length of the pledget. To further minimize chances of failure of the attachment, it may be desired that the attachment mechanism include a longitudinal line of lockstitching 40 in which stitches entirely penetrate the withdrawal cord 12 and the pledget (through both sides), thereby connecting and

affixing the withdrawal cord through a substantial portion of the structure of the pledget, rather than only to an outer surface thereof. Such attachment further diffuses withdrawal force through the body/structure of the pledget. In other examples, a length of withdrawal cord stock may be threaded through a portion of the body/structure of the pledget (e.g., through a hole 45 punched therethrough), looped around and doubled to create pair of trailing portions (not shown). In still other examples, a length of withdrawal cord stock may be looped around a substantial portion of the pledget body without punching, and doubled to create pair of trailing portions (not shown.) The trailing portions of the pair may be tied and knotted or otherwise affixed together. These latter two approaches also may be employed to provide a secure connection between the pledget and the withdrawal cord.

Where lockstitching is used to attach the withdrawal cord to the pledget, it may be desired that the line of lockstitching 40 extend longitudinally along substantially the entire length of the withdrawal cord 12. In examples in which the withdrawal cord 12 is formed of a section of twisted, braided or knitted strands or fibers, lockstitching that traverses substantially the entire length of the withdrawal cord may be desired because the thread strands forming the stitches through the cord are effectively intertwined with component fibers and/or strands of the cord, and can thereby function to substantially prevent the cord from unraveling from its cut ends. Herein, a line of "lockstitching" means a line of stitches formed of at least two strands of thread 40a, 40b disposed on opposing sides of the body(ies) to be stitched together, wherein stitches are sequentially formed as each thread meets and loops around the other, via passage through the body(ies) by one or both threads, at suitable intervals corresponding to the desired size of the stitch. In some examples, a first thread may be sequentially passed through the body(ies) to meet the second thread via use of an appropriate sewing needle, while the second thread is looped about the first thread by operation of a looper. Chainstitching consisting of two threads as described above is included within the definition. A non-limiting example of lockstitching may be seen in FIGS. 4A and 4B, depicting a longitudinal cross section through a wicking member 15 and withdrawal cord 12, wherein these two components are held together by a longitudinal line of lockstitching with stitches formed by front thread 40a and rear thread 40b. FIGS. 4A and 4B depict ISO #301 type lockstitching, as specified by the International Organization for Standardization, ISO 4519:1991, as an example. Other types of lockstitching may be preferred in some circumstances, for example, ISO #401 type chainstitching, which may further enhance the stitches' ability to resist unraveling themselves, and prevent unraveling of the stitched withdrawal cord and/or wicking member, at cut forward and/or rearward ends thereof.

In some circumstances in which the line of lockstitching exists along a length of the trailing portion 12b of the withdrawal cord, it may be desired that the threads used to form the line of lockstitching be made of a suitably hydrophobic fiber material, or fiber material treated to be suitably hydrophobic, so that the lockstitching thread is unlikely to wick fluid along the trailing portion of the withdrawal cord. In some examples the lockstitching thread may be formed of or include cotton fiber, processed or treated to be suitably hydrophobic. In some examples the lockstitching thread may be formed of or include polyester fiber (which in some formulations may be inherently somewhat hydrophobic). In some examples the lockstitching thread may be formed of or

include a blend of cotton fiber and polyester fiber, wherein the cotton fiber may be processed or treated to be suitably hydrophobic.

The tampon **10** may also be provided with multiple withdrawal cords **12**. For example, two withdrawal cords **12** may be attached down the length of the pledget **11** and extend from the withdrawal end thereof. In such an instance, the wicking member, may be joined to one or both of the withdrawal cords **12**.

Especially when the wicking member **15** is joined to the withdrawal cord **12**, the withdrawal cord **12** is preferably non-absorbent along at least the location of such attachment. As used herein, the term “non-absorbent” refers to a structure formed predominately of suitably hydrophobic materials such it does not tend to attract, wick or retain any substantial quantity of aqueous fluid within its structure. In some examples it may be desired that substantially the entire withdrawal cord **12** be hydrophobic, so that the withdrawal cord does not wick menstrual fluid along its trailing portion **12b**, potentially out to its trailing end. The materials comprising the withdrawal cord may be inherently non-wettable or hydrophobic, or they may be treated to provide such properties. For example, a suitable wax may be applied to the withdrawal cord **12** to decrease or eliminate wicking tendency. Other means for providing a material suitable for use as a withdrawal cord **12** which is non-absorbent and/or non-wicking are known in the art. For example, U.S. Pat. No. 5,458,589 describes one such approach. However, the withdrawal cord **12** need not necessarily be non-wicking along its entire length, even if a non-absorbent withdrawal cord is desired. For example, it may be desirable to provide a withdrawal cord **12** in which at least a portion of the cord has a tendency or capability to wick deposited fluid upwardly toward the rearward end **17** of the pledget and into the structure thereof.

The withdrawal cord **12** need not have uniform properties throughout its length. For example, the portion of the withdrawal cord nearest the pledget **11** may be manufactured and/or treated so as to have wicking capability, while the lower portion (i.e. furthest from the pledget **11**) of the withdrawal cord **12** may be manufactured and/or treated so as to not have wicking capability. Other properties such as hydrophilicity/hydrophobicity, density, capillary size, width, thickness, and the like may also vary along the length of the withdrawal cord **12**.

The withdrawal cord **12** may be formed of a strand or strands of component yarn or thread material. In some examples the yarn or thread material may be formed of cotton fiber, cotton fiber processed or treated to be suitably hydrophobic, other natural plant-based fiber which may be processed or treated to be suitably hydrophobic, or polyester, or a combination or blend thereof.

The component yarn or thread may be knitted, twisted or braided to form the withdrawal cord stock. For maximized tensile strength per unit decitex of the withdrawal cord stock, it may be desired that the component yarn or thread be of twisted or braided construction (rather than of knitted, woven or other construction).

Tampons of the type and configurations contemplated herein may also have or include any combination of features described in U.S. Application Ser. No. 62/780,388, filed on Dec. 17, 2018 by Strong et al. and/or U.S. Application Ser. No. 62/834,427, filed on Apr. 16, 2019 by Strong et al.

Wicking Member

The tampon **10** also may be provided with a wicking member **15**. A leading portion **15a** of the wicking member may be attached along a portion or length of the pledget **11**

and may have a trailing portion **15b** extending or trailing by a suitable length from the rearward end **17** of the pledget. The wicking member **15** may be separate from, or joined to, the withdrawal cord **12** along a portion or all of their respective lengths. As will be discussed below, the wicking member **15** may be provided, following insertion of the tampon at a suitable location within the vaginal cavity, to extend rearward of the rearward end of the pledget, further down the vaginal cavity toward the introitus, where it can be in position to contact menstrual fluid that may be present below the pledget, and attract and wick liquidous components thereof up to the pledget **11**.

The wicking member **15** may be formed of a suitable configuration of fibrous material having suitable fluid handling properties and tensile strength. It may be desired that the wicking member be formed separately of the withdrawal cord, and of one or more material(s) distinct from the one or more materials forming the withdrawal cord. As discussed above, in some examples it may be desired that the material(s) forming the withdrawal cord be suitably hydrophobic, to reduce or avoid wicking of fluid along the withdrawal cord. This is contraindicated by the requirements for the wicking member as discussed herein, and it may be desired that hydrophobic fibrous components of the withdrawal cord not be present within the structure of the wicking member, where they can contribute to obstructing or interrupting wicking. Accordingly, it may be desired that the respective structures of the wicking member and withdrawal cord not be coaxial and/or not intermingled. Rather, as suggested in the figures, the wicking member and the withdrawal cord may be arranged in contact with each other (or not) along a substantially parallel, non-coaxial configuration; see, e.g., FIG. 3B, depicting such a non-coaxial configuration, where the parallel longitudinal axes **100a**, **100b** of the wicking member **15** and withdrawal cord **12**, respectively, are not the same.

The ability and tendency of a fibrous structure to draw in and transport (herein, “wick”) aqueous fluid against the influence of external forces acting on the fluid (such as gravity) is a function of several features of the structure. These include the extent of hydrophilicity of the surfaces of the fibers; the extent of capillarity within the structure (where capillarity relates to the number and average size and volume of interstitial spaces constituting potential fluid passageways between and among the fibers, resulting from the extent and manner of fiber consolidation in the structure); the complexity of the fibers’ surface geometry(ies); and the extent to which the structure has already drawn in and retains (i.e., has absorbed) fluid (i.e., saturation level). Capillarity of a fibrous structure relates to the amount of fiber surface area that is present within the structure, per unit volume of the overall structure, and to the density of consolidation of the fibers in the structure, which affects the size and volume of the interstitial spaces or fluid passageways. The size and volume of the interstitial passageways affect the degree to which the aggregate attractive pull of the hydrophilic fiber surfaces in contact with the fluid can overcome forces that resist it, i.e., surface tension of the fluid and external forces such as, e.g., gravity or pressure differential. For example, for a structure formed of a given fiber composition, interstitial passageways which are too large can make the structure ineffective at wicking upward against gravitational pull because there is an insufficient aggregate area of hydrophilic fiber surfaces in contact with the fluid to create attractive pull sufficient to overcome gravitational pull acting on the relatively large fluid volume and mass in the relatively large passageways, and surface tension of the

fluid mass itself tending to resist separation into smaller fluid volumes. On the other hand, interstitial passageways which are too small and/or insufficient in aggregate volume such that, while effective at moving fluid in small volume, can be physically restrictive with respect to wicking volume flow rate. For a given type of hydrophilic fibers, there will be an optimum capillarity in a structure formed of them, at which wicking potential is maximized.

Further, the composition of human menstrual fluid differs from pure water or typical saline test solution in ways (e.g. surface tension) that cause the fluid to behave differently than water or saline solution with respect to wicking structures and absorbent structures. For this reason, a given structure may more readily and/or rapidly wick or absorb a greater quantity of water or saline than menstrual fluid, and vice versa. For purposes herein, wicking and absorption of human menstrual fluid, relevant compositional aspects of which are herein deemed suitably approximated by defibrinated sheep's blood under conditions described herein, are of interest and focus. For a structure formed of a given fiber composition there will be a level of fiber consolidation that optimizes capillarity and wicking performance under these conditions.

Wicking and absorption are relatively complex phenomena and can be difficult to precisely measure and characterize for many types of structures. However, it may be observed, generally, that: (1) between two dry fibrous structures formed of identically composed fibers having similar hydrophilicity, the structure with the more optimal capillarity will have the greater wicking performance; (2) between two dry fibrous structures formed of fibers of differing compositions but having similar capillarity, the structure formed of the fibers having greater hydrophilicity will have the greater wicking performance; (3) between two wetted fibrous structures having similar capillarity, formed of fibers of the same composition and having similar hydrophilicity, the structure holding the lesser quantity of aqueous fluid per unit structure volume will have the greater wicking potential. Between two differing first and second fibrous structures that are placed in contact with each other, the first structure will draw fluid from the second structure if the first structure has a combination of hydrophilicity, capillarity and level of fluid content (saturation level) per unit structure volume that impart to it greater wicking potential than the second structure. Conversely, if the first structure has lesser wicking potential than the second structure, the first structure will not draw fluid from the second structure.

Thus, it may be desired that the component fibrous material of the wicking member 15 have a combination of fibers with hydrophilic surface properties, and optimized capillarity, to promote wicking of fluid therealong, but at the same time, not have, or not be assembled in a configuration having, a combination of capillarity and hydrophilicity that render it more likely to attract and retain menstrual fluid against the wicking potential of the material(s) of the pledget 11. It is preferred that the material of the wicking member serve to wick fluid to the pledget, but that the material(s) of the pledget have greater wicking potential and absorbency so as to be effective at drawing menstrual fluid from the wicking member over the expected normal duration of use of the tampon.

In some examples the wicking member may be formed of fibrous material(s) similar to those used to form the pledget, e.g., rayon fibers, absorbent cotton fibers or any combination thereof. In such examples the fibrous materials forming the wicking member and pledget should be configured such that the body of the wicking member has less wicking potential

than the pledget. The wicking potential of the wicking member can be adjusted by the selection of material of which it is formed, for its relative level of fiber surface hydrophilicity and its relative capillarity. Capillarity may be adjusted by the manner in which the fibers forming the wicking member are consolidated and densified within the structure.

It has been discovered that a non-hydrophilic (i.e., hydrophobic) fibrous material can be effectively treated and configured to serve the desired wicking function while in most circumstances having less affinity for the fluid than typical pledget materials (i.e., rayon fiber and/or cotton fiber). Spun polymeric synthetic fibers that are ordinarily hydrophobic, such as spun polypropylene fibers, may be treated, e.g., via application of a suitable surfactant finish, to render their surfaces hydrophilic. Since such fibers typically have simple and/or smooth and non-porous surface geometry, however, the fiber surfaces themselves do not substantially contribute to capillarity, and bundles or assemblies of such fibers will have substantially less wicking potential (and tendency to retain the fluid) than bundles or assemblies of more complexly-shaped hydrophilic fibers such as rayon, cotton, or other natural plant-based fibers. Accordingly, in some examples it may be desired that the wicking member be formed of fibrous material including spun polypropylene or other spun polymeric synthetic polymer.

For a more detailed description of hydrophilicity and contact angles see the following publications which are incorporated by reference herein: The American Chemical Society Publication entitled "Contact Angle, Wettability, and Adhesion," edited by Robert F. Gould, and copyrighted in 1964; and TRI/Princeton Publications, Publication Number 459, entitled "A Microtechnique for Determining Surface Tension," published in April 1992, and Publication Number 468 entitled, "Determining Contact Angles Within Porous Networks," published in January, 1993, both edited by Dr. H. G. Heilweil.

While a tampon with a wicking member may absorb some menstrual fluid into the wicking member and may even wick fluid to the pledget to some extent, it is believed from research that the effectiveness of the combination may not be meaningfully noticeable to a user unless its ability to capture and wick fluid through the wicking member up to the pledget exceeds a particular value for Wicking as set forth and described herein. A combination of materials described herein, used to constitute and configure a tampon product, may be selected and assembled as described to provide a tampon that will wick at least 1.2 grams, more preferably at least 1.5 grams and even more preferably at least 3.2 grams of test fluid up through the wicking member as measured using the Wicking Measurement method herein. Information herein and also as known in the art is sufficient to enable one to select materials for the pledget and for the wicking member to achieve these levels of wicking. If the material of the wicking member has an insufficient combination of suitable hydrophilicity and capillarity, it will be unable to attract and wick menstrual fluid upwardly to the pledget to the levels specified herein, under the conditions of the measurement method (which are designed to approximate the orientation of the tampon and pressure to which its materials are subjected when the tampon is in use, disposed in the vaginal cavity). For this reason, a wicking member formed of polymeric fibers, for example, that have not been suitably processed or treated to render them suitably hydrophilic, have insufficient longitudinal directional orientation, and/or are too loosely or too densely consolidated, will be ineffective. On the other hand, if the material of the wicking

13

member has a combination of hydrophilicity at a suitable level and capillarity that makes it have a greater affinity for fluid contained therein than can be overcome by the wicking potential of the pledget, the pledget will be unable to draw fluid away and out of the wicking member, and once saturated, the wicking member will cease wicking. For this reason, a wicking member formed primarily of, for example, completely scoured cotton fibers, and/or rayon fibers (which have a high affinity for aqueous fluid and therefore form structures that are relatively highly absorbent), may be unsatisfactory. On the other hand, a blend comprising, for example, no greater than about 75 weight percent, more preferably no greater than about 63 weight percent, and even more preferably no greater than about 50 weight percent cotton, rayon (or viscose, or lyocell), or any combination thereof, with the balance constituted by synthetic fibers such as polyethylene, polyethylene terephthalate (PET), polypropylene, polyester, polyamide, or any combination thereof, may impart suitable wicking properties but still yield fluid up to a pledget formed predominantly of cotton, rayon (or viscose, or lyocell), or a combination thereof, and suitably structured. The ability of the pledget to draw fluid from the wicking member may be further enhanced by processes and configurations described in U.S. provisional patent application Ser. No. 62/683,661. A balance between wicking potential of the pledget and wicking potential of the wicking member may be identified to meet the wicking levels specified above. It has been learned that a level of measured wicking at one or more levels specified above is greater than that achieved by currently available tampons that include wicking structures. It is believed that a level of measured wicking at levels specified above represents improvement in the performance of tampons with wicking structures, in preventing bypass leakage or leakage of residual fluid in the vaginal cavity present following removal of a used tampon. Using combinations of materials described herein, the inventors have achieved measured wicking as high as 3.5 grams, although it is contemplated that greater levels as high as 3 grams, 4 grams or even 5 grams may be achievable through experimentation, using suitable combinations of materials and configurations identified herein or otherwise known to the person of ordinary skill in the art.

In order that the wicking member provide continuous, substantially uninterrupted and generally longitudinally-directed pathways for relatively rapid fluid travel along the length of the wicking member, it may be desired that the configuration of fibrous material be formed of carded, directionally-biased spunbond or otherwise directionally biased fibers, with fibers predominately biased in the longitudinal direction of the tampon.

In some examples, the wicking member may have the form of a strip of nonwoven, woven or knitted cloth, or ribbon. However, for purposes of maximizing the presence of generally longitudinally-oriented fluid pathways for wicking fluid to the pledget, it may be desired that the material forming the wicking member be a nonwoven web material, and that fibers forming the material are predominately longitudinally oriented or biased, and held together by discontinuous thermal calender bonding, comprising a pattern of discrete bonds constituting a bonded area of no more than about 20 percent, more preferably no more than about 15 percent, even more preferably no more than about 12 percent, and still more preferably no more than about 10 percent. (Herein, "bonded area" percentage is the percentage of surface area of a nonwoven web material on one side, that is occupied by bonds. Bonded area of a calender-bonded nonwoven web material is typically substantially reflected in

14

the bonding area of the radially outermost surfaces of bonding pins on a calender bonding roller used to bond the material.) Fibers that are predominately longitudinally oriented or biased may better serve to create the presence, number and length of generally longitudinally-biased pathways through and along the fibrous material forming the wicking member, along which fluid may travel more directly along hydrophilic surfaces thereof toward the pledget.

A single layer or ply of web material having a suitable basis weight may be used to form wicking member 15. However, it has been discovered that a multi-ply/multi-layer structure that may be formed by folding a relatively lower basis weight web material over on itself, along longitudinal fold lines, in a manner that appears to increase wicking efficacy as compared with a single-layer structure having a basis weight comparable to that of the multi-layer/multi-ply structure. Without intending to be bound by theory, it is believed that the several plies or layers in folded configuration provide added, alternative fluid passageways with regions having variations in size and capillarity, such that alternative fluid passageways are presented around regions in which wicking may be impaired by, e.g., regions of excessively high or low fiber density/compaction. Further, it has been observed that folding such material from a rolled supply may be accomplished efficiently on a manufacturing line, using suitable folding plows or folding tables. The web material may be a woven or nonwoven web material; in some circumstances a nonwoven web material having a longitudinal fiber direction bias may be preferred, for reasons discussed above.

Referring to FIGS. 5A-5F, the wicking member 15 may be formed of a section of web material that is folded to form two or more plies/layers, along one or more longitudinal fold lines 15c, which may be disposed proximate to longitudinal side edges of the wicking member. It is believed that three plies/layers may wick more effectively than two, and four may wick more effectively than three. Accordingly, three or more, or even four or more, plies/layers may be preferred. The folding arrangement may take the form of a "V" configuration (FIG. 5A); a "Z" configuration (FIG. 5B); a "roll" configuration (FIG. 5C); a "W" configuration (FIG. 5D); or "C" configuration (FIG. 5E); or a combination of a roll configuration and a C configuration (FIG. 5F). It may be desired that the folding configuration selected will enclose as many longitudinal edges of the web material as possible within the fold structure, to keep the number of exposed longitudinal material edges 15d to 1, or 0, to minimize the potential for fraying at cut edges and impart the wicking member with a more finished appearance. Thus, folding configurations such as those depicted in, for example, FIGS. 5C, 5E and 5F may be preferred because no more than one edge 15d of the material is exposed/not enclosed.

It will be appreciated that FIGS. 5D and 5E reflect a wicking member 15 formed of a section of web material having three longitudinal folds, resulting in four layers of material present along a longitudinally central portion of the wicking member. In a similar fashion, FIG. 5E reflects that a wicking member may be formed with four longitudinal folds along longitudinal fold lines 15c, resulting in five layers of material present through a longitudinally central portion of the wicking member. In some circumstances, it may be desired that the number of layers of material present in the wicking member 15 along a longitudinally central portion of the wicking member not exceed six layers (e.g., imparted by five folds), more preferably, not exceed five layers (e.g., imparted by four folds), and even more preferably, not exceed four layers (e.g., imparted by three folds);

15

this is to avoid making the wicking member too bulky for a conventional applicator and potentially uncomfortable for the user.

When the wicking member is formed of fibrous material as described herein, and has cut forward and/or rearward ends **15e**, **15f** (FIGS. 2, 3B), it may be desired that the fibrous material at one or both of forward and rearward ends be consolidated and bound together (“finished”) via a mechanism that effectively reduces chances of the occurrence, or reduces the extent to which, the fibrous material may fray, unravel or dislodge at the cut ends. In non-limiting examples, in which the wicking member includes or is formed of fibrous material manufactured from thermoplastic resin, fibers proximate one or both the forward and rearward ends of the wicking member may be fused, welded or otherwise bonded together by, e.g., application of heating energy and/or pressure during or following cutting. It will be appreciated that other mechanisms may serve, such as, but not limited to, knotting the cut end, applying a binder or adhesive to the cut end, etc.

The wicking member **12** may have any suitable length, but its trailing portion **15b** (the portion extending rearward of rearward end **17** of pledget **11**) is preferably shorter than the trailing portion of the withdrawal cord **15**. The trailing portion of the wicking member should not be long enough to extend through the introitus when the tampon **10** is fully inserted and properly positioned within the vaginal cavity. Although dimensions of the vaginal cavity vary among individual users, it may be desired for most users that the trailing portion **15b** of the wicking member **15** should have a trailing length TL no greater than about 60 mm, more preferably no greater than about 50 mm, even more preferably no greater than about 40 mm, and still more preferably no greater than about 30 mm, and preferably no less than about 10 mm, more preferably no less than about 15 mm, and even more preferably no less than about 20 mm. (For purposes herein, the length of the trailing portion is measured with the trailing portion held in a straightened position, but in a relaxed condition, i.e., not under longitudinal tension greater than about 5 gf).

In order to ensure that an adequate portion of the surface area of the wicking member **15** is exposed to contact with the pledget (for purposes of facilitating movement of fluid from the wicking member to the pledget), it may be desired that the leading portion **15a** have a length that is at least one-quarter of the total length PL of the pledget, and more preferably at least one-third of the total length PL of the pledget. In a particular example, the leading and trailing portions of the wicking member **15a**, **15b** may be approximately equal in length. It may also be preferred that the leading portion of the wicking member be affixed to the pledget along a length that is at least 10 mm, and more preferably at least 15 mm.

To enhance unitized structural integrity of the tampon, it may be desired that the wicking member **15** be lockstitched to the pledget, in a manner such as described above. In some examples the withdrawal cord **12** and the wicking member **15** may be lockstitched together to the pledget, via the same configuration/line of lockstitching. In such configuration, it may be desired that the wicking member be disposed in direct contact with the pledget, and preferably that the wicking member be disposed between the pledget and with the withdrawal cord, to provide for direct contact and fluid transfer between the wicking member and the pledget, unobstructed by the (e.g., hydrophobic) structure of the withdrawal cord. In some examples, however, the wicking member **15** may be attached to the pledget **11** by a mecha-

16

nism differing from that attaching the withdrawal cord **12** to the pledget **11**, and may also be physically separated from the withdrawal cord **12** at or along location(s) of attachment on the pledget. In some examples, the wicking member **15** may be attached to the pledget **11** by adhesive bonds, by thermal compression or ultrasonic bonds (in which respective material(s) of the wicking member and of the pledget are fused or welded together) or may be stitched to the pledget by stitches separate from stitches attaching the withdrawal cord to the pledget **11**.

In some further examples, one or more threads used to stitch the wicking member **15** to the pledget **11** may be selected for suitable tensile strength and hydrophobicity such that they may be extended beyond the trailing rearward end of the wicking member by a suitable length, and by themselves serve as the withdrawal cord. This configuration eliminates the need for, and expense and complexity associated with including, a separate withdrawal cord. Where two or more stitching threads are used to lockstitch the wicking member to the pledget, they may be twisted, braided or otherwise suitably intertwined or combined into a singularized cord configuration in a trailing portion extending rearwardly from the wicking member.

It may be desired that the material used to form the wicking member be tinted or pigmented to impart the wicking member with a color that visibly contrasts with the color(s) of the materials forming the pledget and/or withdrawal cord. This may be deemed useful for visually signaling to the user that a differing material is present in the wicking member, suggesting a functionality distinct from that of the withdrawal cord. In connection with appropriate information on, or associated with, packaging for the tampon product, such tinting or pigmentation can advantageously serve to remind the user that the wicking member is present to provide supplemental protection against leakage. When the pledget and/or materials constituting the pledget have a substantially white color (which is, for example, the natural color for suitably processed, undyed cotton, and is believed to be preferred by many consumers because it connotes purity, cleanliness, sanitation and/or freshness), it may be desired that the material(s) constituting the wicking member are imparted with a non-white color that not only visibly contrasts with the color of the pledget, but also visibly contrasts with color(s) of areas of a pledget as stained by menstrual fluid as it may appear immediately following use. Thus, in some examples it may be preferred that the non-white color of the wicking member be selected from a range of colors that will visibly contrast with the color of the pledget, and with the color(s) of the pledget when stained by menstrual fluid, as it appears immediately following withdrawal of the tampon. This coloration feature may serve to provide the user with additional signal of the functionality of the wicking member. For purposes herein, “substantially white” means having CIE L*a*b* values when measured according to the Color Measurement Method set forth below, in which L* is ≥ 87 , and the absolute values of each of a* and b* are ≤ 2 . A wicking member color that visibly contrasts with the color of the pledget is any color that exhibits a $\Delta E^* \geq 15$ from the color of the pledget, measured according to the Color Measurement Method below. Techniques for imparting varying colors, and adjusting the depth thereof, to synthetic, semi-synthetic and natural plant-based fibers or filaments, or materials made therefrom (e.g., via use or inclusion of pigments, dyes or inks), are known in the art.

In an array of two or more packaged tampon products (herein “array” means two or more differing products of the same brand, marketed or appearing for sale simultaneously

17

in the same or proximate respective locations (physical or online/virtual) (e.g., on the same or respective proximately located shelves within the same retail store)), differing wicking members can be included with tampons of differing features, for the purposes of functioning differently with the differing tampon products, signaling the differences in features to consumers, or a combination of both. In one non-limiting example, packaged tampons having a first absorption capacity may include wicking members imparted with a first color, while packaged tampons having a second, differing absorption capacity may include wicking members imparted with a second color visually distinguishable from the first color. For purposes herein, a second wicking member color is “visually distinguishable” from a first wicking member color when the second wicking member color exhibits a $\Delta E^* \geq 5$ from the first wicking member color, measured according to the Color Measurement Method below. Wicking members for an array of differing tampon products may differ not only in color, but in other characteristics such as material composition and/or absorbency/wicking characteristics, physical structure (e.g., braided, twisted, knitted, etc.), length, width, diameter, density, decitex or other dimension, location of attachment on the pledget, etc. The differing tampon products with respectively differing wicking members may be accompanied by associated packaging material imprinted with graphic/pictorial information, verbal information, or a combination thereof, that signals the differences in the respective products and/or wicking members.

A tampon as contemplated herein is believed to offer several advantages over prior art tampons. As noted previously, the incorporation of the wicking member **15** extends fluid capturing capability to lower regions of the vaginal cavity. Additionally, because the tampon can be manufactured by processes in which the wicking member is less compressed than the pledget **11**, the material forming the wicking member can be available to immediately draw in fluid, without the need for re-expansion from a compressed state.

Tampons of the type and configuration(s) contemplated herein may be manufactured via the process described in U.S. Application Ser. No. 62/780,388, filed on Dec. 17, 2018 by Strong et al. and/or U.S. Application Ser. No. 62/834,427, filed on Apr. 16, 2019 by Strong et al.

To form a tampon ready for use, the tampon pledget **11** may be compressed and heat conditioned (which may include use of steam or elevated humidity) in any suitable conventional manner to impart it with a self-sustained form suitable for easy and comfortable insertion, which may be a cylindrical form. Pressures, temperatures and humidity conditions suitable for this purpose are known in the art. Typically, the pledget **11** is compressed in both the radial and axial direction using any suitable means known in the art. While a variety of techniques are known and acceptable for these purposes, a modified tampon compressor machine available from Hauni Machines, Richmond, VA, is suitable.

The tampon **10** contemplated herein may be inserted digitally or via the use of an applicator. If the tampon **10** is to be configured for digital insertion, or for insertion from a generally cylindrical applicator, it may be desirable to form the pledget from a layer of absorbent material which has been rolled or otherwise formed into a cylindrical or capsule shape.

Any of the currently available tampon applicators may also be used for insertion of the tampon contemplated herein. Such applicators of typically a tube-and-plunger type arrangement and may be plastic, paper, or other suitable

18

material. A compact type applicator can also be suitable. The applicator plunger may be depressed by the user to push the compressed pledget **11** out of the applicator while fitting around the wicking member **15**.

5 Color Measurement

The total color difference (ΔE^*) between a tampon pledget **11** and its wicking member **15** is calculated from the $L^* a^* b^*$ color values obtained for each respective portion of the tampon. Color analyses are made using a $0^\circ/45^\circ$ spectrophotometer with adjustable apertures capable of making standard CIE $L^* a^* b^*$ measurements in accordance with ASTM E1349. An example of a suitable spectrophotometer is the Labscan XE (available from Hunter Associates Laboratory, Inc., Reston, VA, or equivalent). All testing is performed in a room maintained at a temperature of $23^\circ \text{C} \pm 2.0^\circ \text{C}$. and a relative humidity of $50\% \pm 2\%$ and samples are conditioned under the same environmental conditions for at least 2 hours prior to testing.

If the tampon is provided in an application, the test sample is prepared by first removing the tampon **10** from the applicator in the manner the product is designed to effect ejection of the tampon **10** from the applicator. The pledget **11** is flattened out by gently opening it from its self-sustaining shape. Using a small pair of scissors to sever any stitching as necessary, or using freeze spray to deactivate any adhesive used to join them, gently separate and remove the wicking member **15** from the pledget **11**, using care so as not to damage either component in the process. For each product tested, a total of 5 pledgets and 5 wicking members are prepared in this manner.

To measure color, calibrate and standardize the instrument per the vendor instructions using the standard white and black tiles provided by the vendor. Set the spectrophotometer to use the CIE $L^* a^* b^*$ color space with a D65 standard illumination, a 10° observer, a 0.125 inch area view, a 0.200 inch aperture, and the UV filter set to nominal. Place the pre-flattened pledget test sample over the aperture such that the entire aperture is covered by the pledget **11** on an area free of the withdrawal string **40**. Place the standard white tile behind the pre-flattened pledget test sample, take a reading and record $L^* a^* b^*$ values as $L2^* a2^* b2^*$ to the nearest 0.01 units. Remove the pledget test sample from the aperture and replace it with the wicking member test sample. Ensure that the entire aperture is covered by the wicking member, minimizing the amount of withdrawal cord **12** present in the aperture's viewing area. Place the standard white tile behind the wicking member test sample, take a reading and record $L^* a^* b^*$ values as $L1^* a1^* b1^*$ to the nearest 0.01 units. Calculate the total color difference (ΔE^*) between the pledget and the wicking member as follows:

$$\Delta E^* = [(L2^* - L1^*)^2 + (a2^* - a1^*)^2 + (b2^* - b1^*)^2]^{1/2},$$

and record as ΔE^* to the nearest 0.01 units.

In like fashion, repeat for a total of five measurements obtained on five different tampon pledget and wicking member samples. Calculate the arithmetic mean for ΔE^* obtained from all five measurements and report to the nearest 0.01 unit.

Wicking Measurement

The ability of a tampon configuration with a wicking member to capture fluid in the wicking member, and wick and fluid to the pledget, may be measured using this Wicking Measurement method. A known quantity of test fluid is delivered at a constant rate over a specified amount of time to a portion of the wicking member **15** inside a pressurized wicking chamber. The quantity of fluid absorbed by the tampon **10** is determined and reported as Total Uptake. All

measurement is performed in a laboratory maintained at $23^{\circ}\text{C} \pm 2^{\circ}\text{C}$ and $50\% \pm 2\%$ relative humidity. The measurement equipment as described herein is configured to approximate the pressure to which a tampon is subjected inside the body during actual use.

The measurement apparatus **1000** is schematically depicted in FIG. 9A. The measurement apparatus **1000** includes a wicking chamber **1001** including a hollow glass tube **1001a** mounted to a ring stand **1002** in such a way that the longitudinal axis **1003** of the tube **1001a** lies within a vertical plane and is 30 ± 20 from a vertical line. The glass tube **1001a** is open at both ends, and has a 3.3 cm inner diameter, a 3.8 cm outer diameter, and an overall length of 10.5 cm. The glass tube **1001a** has a $\frac{1}{4}$ inch hose barb **1004** (that provides fluid communication with the space inside the tube) located at its longitudinal midpoint, extending perpendicularly from the outer surface of the glass tube. Located about 7 mm from each end of the glass tube are circumferential sealing ridges **1005a**, **1005b** on the outside of the tube (which may be formed as “mamas,” also known as “mari-
 20 ahs,” as known in glassblowing art), each with a 48.0 mm outer diameter and oriented along a plane perpendicular to the longitudinal axis **1003** of the glass tube. Two lines **1006**, **1007** are marked on the glass tube **1001a**, both oriented along planes perpendicular to the longitudinal axis **1003** of the tube: the Pledget Base Line **1006** is marked at the longitudinal midpoint of the glass tube; the Cannula Position Line **1007** is marked 10.0 mm below the Pledget Base Line **1006**. A compressed air source is connected to the hose barb **1004** with flexible tubing **1008**. The air pressure is controlled via a pressure regulator **1009** and calibrated manometer **1010** (standardized to ANSI standards). The pressure regulator **1009** is set to 0.5 ± 0.02 psi.

An unlubricated condom **1011** (condom complying with ASTM D3492) is installed inside the glass tube **1001a** as follows. Unroll the condom **1011** and mark a Positioning Line thereon that is perpendicular to the longitudinal axis of the condom **1011** and located 12.0 ± 0.1 cm from the open end.

Using a glass rod inserted into the condom **1011**, push the condom **1011**, closed/tip end first, into the lower open end of the glass tube **1001a**, up through the tube and out the upper open end. Cut off the tip of the closed end of the condom **1011** (no more than about 1 cm from the tip) and discard. Align the Positioning Line marked on the condom with the edge of the open upper end of the glass tube **1001a**. Now, stretch and pull the newly cut edge **1020** of the condom **1011** radially out, down and over the circumference of the upper edge of the glass tube **1001a**, and downwardly over and past the upper sealing ridge **1005a**. Secure the circumferential cut edge of the condom **1011** to the outside of the glass tube **1001a** below the upper sealing ridge **1005a**, using a first rubber band **1021**. Now, from the bottom, gently pull the condom slightly taut longitudinally, and stretch the circumferential edge **1022** of the original open end of the condom **1011** radially out, up and over the circumference of the lower edge of the glass tube **1001a** and upwardly past the lower sealing ridge **1005b**, and secure the circumferential edge of the original open end of the condom **1011** to the outside of the glass tube **1001a** above the lower sealing ridge **1005b**, using a second rubber band **1023**.

(The references “up”, “down”, “upper”, “lower”, and similar terms used in this measurement method description are relative the position of the glass tube when mounted to and held by the ring stand as shown in FIG. 9A. However, the condom may be installed within the glass tube before it is mounted on the ring stand. Similarly, the Pledget Base

Line and Cannula Position Line on the glass tube as described above may be marked on the tube before it is mounted on the ring stand.)

The test fluid used (for purposes of providing a fluid having a suitable degree of similarity to human menstrual fluid) is defibrinated sheep's blood, with a packed cell volume between 38%-42% (such as that available from Cleveland Scientific Ltd., Bath, OH, or equivalent) and a viscosity between 6.5-8.0 centistokes. Prior to use in this measurement method, the viscosity of the test fluid is measured using a low viscosity rotary viscometer (a suitable instrument is the Cannon LV-2020 Rotary Viscometer with UL adapter, Cannon Instrument Co., State College, PA, or equivalent). The appropriate size spindle for the viscosity range is selected, and the instrument is operated and calibrated as per the manufacturer's instructions. Measurements are taken at $23^{\circ}\text{C} \pm 1^{\circ}\text{C}$ and at 60 rpm. Results are recorded to the nearest 0.01 centistokes and must be in spec before use.

The test fluid is placed in a 250 mL reservoir with a cover and continuously and moderately stirred to avoid separation. The temperature of the test fluid is maintained at $23^{\circ}\text{C} \pm 2^{\circ}\text{C}$ during use. The test fluid is supplied from the reservoir to a 15-gauge steel laboratory cannula **1012** (about 10 cm long with a blunt tip at each end with an inner diameter of 1.33-1.41 mm and an outer diameter of 1.82-1.84 mm (such as that available from Cadence Science Inc., 2080 Plainfield Pike, Cranston, RI 02921, or equivalent) and cannula sleeve **1013** (transparent flexible silicone tubing with 3.2 mm inner diameter, 4.8 mm outer diameter, about 7 cm long (such as that available from Cole Parmer, Vernon Hills, IL, or equivalent) with peristaltic pump tubing **1014** that has an inner diameter of 1.6 mm. The cannula **1012** is inserted through the prepared cannula sleeve **1013** such that there is about 6 mm of the upper/distal tip of the cannula **1012** extending beyond the upper/distal edge of cannula sleeve **1013**. (The purpose of the cannula sleeve, of an inner diameter greater than the outer diameter of the cannula, is to provide a pathway for fluid that is not captured and wicked after contacting the wicking member, to flow under gravitational pull down and out of the test chamber, thereby helping reduce chances of pooling of the fluid within the test chamber.) The lower end of the cannula **1012** is inserted into the peristaltic pump tubing **1014**. A peristaltic pump **1015** (such as Master Flex, available from Cole Parmer, Vernon Hills, IL, or equivalent) is programmed to deliver $5.0 \text{ g} \pm 0.25 \text{ g}$ of test fluid at $1.0 \text{ g/min} \pm 0.02 \text{ g/min}$ (i.e., over a 5-minute period of operation of the pump). Prior to commencement of the measurement, the peristaltic pump **1015** is calibrated with the test fluid, and the tubing **1014** and cannula **1012** are then primed with test fluid.

A cannula stabilizing bar **1016** is mounted to the ring stand **1002** such that it is about 2 cm below the bottom of the wicking chamber **1001**, however, this position can be adjusted as needed. The cannula stabilizing bar **1016** is used to support the peristaltic pump tubing **1014** in order to allow the cannula **1012** to be maintained in a position in which its longitudinal axis is approximately parallel to longitudinal axis **1003** of the glass tube **1001a** during the measurement.

Tampon measurement samples still in their applicators and wrappers are conditioned at $23^{\circ}\text{C} \pm 2^{\circ}\text{C}$ and $50 \pm 2\%$ relative humidity for at least 2 hours prior to use. Measurement samples are not removed from their wrappers or applicators until immediately prior to use, and must be used within 30 minutes following removal. Clean disposable exam grade, nitrile rubber, powder-free medical gloves must be worn while preparing the measurement samples and

21

during the measurement procedure in order to prevent any contamination from contact with the analyst's hands. Each tampon measurement sample is prepared by first removing the tampon **10** from its applicator in the manner the product is designed to effect ejection of the tampon from the applicator. Cut the withdrawal cord **12** from the tampon **10** at the rear end of the trailing portion **15b** of the wicking member **15**. After cutting away such portion of the withdrawal cord, record the Dry Mass of the tampon measurement sample to the nearest 0.01 g.

Now referring to FIG. **9B**, the tampon measurement sample is placed into the unpressurized wicking chamber **1001** as follows. (A second analyst may be required to assist in manipulating the measurement sample and cannula into position and holding them in position until the measurement chamber is pressurized.) Insert the forward end **16** of the pledget **11** into the bottom of the wicking chamber **1001** and move the sample upward in the chamber to a location at which the rearward end **17** of the pledget **11** is aligned with the Pledget Base Line **1006**. Align the longitudinal axis of the pledget and wicking member **15** approximately with the longitudinal axis **1003** of the wicking chamber **1001**. If the wicking member **15** and remaining portion of withdrawal cord **12** are not coaxial, rotate the tampon measurement sample about its longitudinal axis within the chamber, to a position in which the trailing portion **15b** of the wicking member **15** occupies a predominately dorsal (overlying) position relative the remaining portion of the withdrawal cord, and the remaining portion of the withdrawal cord **12** occupies a predominately ventral (underlying) position relative the trailing portion **15b** of the wicking member. While holding the measurement sample in place inside the wicking chamber **1001** in this position, insert the prepared cannula **1012** with attached cannula sleeve **1013** into the bottom of the wicking chamber **1001** and move it upward within the chamber to a position at which the upward tip of the cannula **1012** is aligned with the Cannula Position Line **1007**, and positioned directly over and in contact with the dorsal surface of the wicking member **15**, and overlaps it by at least 5 mm. (If the trailing portion **15b** of wicking member **15** of the particular tampon sample is shorter than 15 mm when in a straightened but substantially relaxed condition so as to provide ≥ 5 mm overlap, position the cannula tip over the wicking member so that it overlaps the trailing end of the wicking member by $\frac{1}{3}$ of the length of the trailing portion **15b**. If the trailing portion **15b** of wicking member **15** of the particular tampon sample is shorter than 6 mm when in a straightened but substantially relaxed condition, then the sample cannot be tested according to this method, and does not fall within the contemplation of claims herein that recite a value for Wicking.)

Now, pressurize the wicking chamber **1001** to 0.5+0.02 psi (over ambient air pressure) such that the condom **1011** inflates within the wicking chamber around the measurement sample and cannula **1012** to hold them in place inside the chamber **1001**. Ensure that there is still a distance of 6 mm between the tip of the cannula **1012** and the edge of the cannula sleeve **1013**. Adjust the position of the cannula stabilizing bar **1016** so that it supports the peristaltic pump tubing **1014** to allow the longitudinal axis of the cannula **1012** to be maintained in a position that is approximately parallel to the longitudinal axis **1003** of the wicking chamber **1001**. Adjust as needed to maintain this position throughout the measurement.

Prior to starting of the peristaltic pump, a tray **1017** or other suitable collection means may be placed on the benchtop below the bottom of the cannula sleeve **1013** and

22

wicking chamber **1001** to collect any test fluid that is not wicked/absorbed by the measurement sample and exits the bottom of the test chamber.

Start the peristaltic pump **1015** to deliver 5.0 g+0.25 g of test fluid at 1.0 g/min $\pm$ 0.02 g/min through the cannula **1012**. While the test fluid is being delivered, ensure that the tip of the cannula **1012** remains in contact with the trailing portion **15b** of the wicking member **15** and that there is no pooling of fluid in a crease in the condom **1011** proximate the cannula tip. If pooling of test fluid is observed, move the cannula **1012** slightly further down the trailing portion **15b** of the wicking member **15**. If pooling of test fluid continues, discard the tampon and repeat using a new measurement sample.

After the pump stops, depressurize the wicking chamber **1001** and remove the test sample. Record the Wet Mass of the test sample to the nearest 0.01 g. Calculate the mass of fluid absorbed by the test sample as Wet Mass—Dry Mass and record as Total Uptake to the nearest 0.01 g. The condom **1011** is wiped clean in between test samples and replaced after every 10 samples are tested.

In like fashion, repeat for a total of ten replicate test samples. Calculate the arithmetic mean for Total Uptake measured across all ten replicate test samples and report to the nearest 0.01 g.

In view of the description above, the following non-limiting examples of combinations of features in tampons are contemplated:

1. A tampon (**10**), comprising:
 - a pledget (**11**), when in an opened configuration, having a first broad surface (**35**) and second broad surface (**36**) opposite the first broad surface, each of the first and second broad surfaces being defined by a forward end edge (**16**), a rearward end edge (**17**), and a pair of opposing longitudinal edges (**37a**, **37b**); the first broad surface having a pledget width PW measured between the longitudinal edges; the pledget comprising a mass of fibrous absorbent material (**31**) disposed within or between one or more layers of an outer cover (**30**), the outer cover being formed of a first nonwoven web material;
 - a withdrawal cord (**12**) joined to the pledget and trailing from the rearward end edge, and
 - a wicking member (**15**) comprising a second nonwoven web material that is distinct from the first nonwoven web material and distinct from the withdrawal cord, the wicking member being joined in direct contact with the first broad surface (**35**), the wicking member having a wicking member width MW that is less than the pledget width; the wicking member having a trailing portion (**15B**) extending rearwardly from the rearward end of the pledget by a trailing length.
2. The tampon of example 1 wherein the wicking member comprises at least two plies.
3. The tampon of example 2 wherein the at least 2 plies both comprise the second nonwoven web material.
4. The tampon of either of examples 2 or 3 wherein the at least 2 plies result from folding the second nonwoven web material along at least one longitudinal fold line.
5. The tampon of any of the preceding examples wherein the wicking member has at least 3 plies and no more than 1 exposed longitudinal material edge.
6. The tampon of example 5 wherein the second nonwoven web material is folded along longitudinal lines to form at least 3 plies in a roll fold configuration.

23

7. The tampon of example 5 wherein the section of second nonwoven web material is folded along longitudinal lines to form at least 4 plies in a "C" fold configuration.
8. The tampon of any of the preceding examples wherein the wicking member is stitched to the pledget.
9. The tampon of either of any of the preceding examples wherein the wicking member and the withdrawal cord are stitched to the pledget.
10. The tampon of example 9 wherein the wicking member and the withdrawal cord are stitched to the pledget at least in part by common stitches (40).
11. The tampon of any of the preceding examples wherein the second nonwoven material has a fiber directional bias along a longitudinal direction.
12. The tampon of any of the preceding examples wherein the second nonwoven web material is a spunbond nonwoven web material.
13. The tampon of example 12 wherein the spunbond nonwoven web material is bonded in a bond pattern wherein bonds in the bond pattern occupy no more than 20 percent, more preferably no more than 15 percent, and even more preferably no more than 12 percent of the surface area of the spunbond nonwoven web material on one side.
14. The tampon of either of any of examples 11-13 wherein the second nonwoven web material has a basis weight of 10-40 gsm.
15. The tampon of any of examples 1-11 wherein the second nonwoven web material is a carded spunlace nonwoven web material.
16. The tampon of any of the preceding examples wherein the second nonwoven web material comprises fibers selected from the group consisting of polyester fibers, polyethylene fibers, PET fibers, polypropylene fibers, cotton fibers, rayon fibers, viscose fibers, lyocell fibers, and any combination thereof.
17. The tampon of example 16 wherein the second nonwoven web material comprises no more than about 75 weight percent, more preferably no more than about 63 weight percent, and even more preferably no more than about 50 weight percent, of fibers selected from the group consisting of cotton fibers, rayon fibers, viscose fibers, lyocell fibers, and any combination thereof.
18. The tampon of example 16 wherein the second nonwoven web material comprises predominantly synthetic fibers.
19. The tampon of example 16 wherein one or more components of the second nonwoven web material have been treated and thereby rendered hydrophilic.
20. The tampon of any of the preceding examples wherein the second nonwoven web material comprises cotton.
21. The tampon of example 20 wherein the cotton has been left at least partially unstripped of natural waxes and oils and/or has been treated and thereby reduced in hydrophobicity.
22. The tampon of any of the preceding examples wherein material comprised by the withdrawal cord is hydrophobic.
23. The tampon of any of the preceding claims that exhibits wicking of at least 1.2 grams, more preferably at least 1.5 grams and even more preferably at least 1.8 grams, up to 3.2 grams, more preferably up to 4 grams, and even more preferably up to 5 grams.

The dimensions and values disclosed herein are not to be understood as being strictly limited to the exact numerical values recited. Instead, unless otherwise specified, each such dimension is intended to mean both the recited value and a

24

functionally equivalent range surrounding that value. For example, a dimension disclosed as "40 mm" is intended to mean "about 40 mm."

Every document cited herein, including any cross referenced or related patent or application is hereby incorporated herein by reference in its entirety unless expressly excluded or otherwise limited. The citation of any document is not an admission that it is prior art with respect to any invention disclosed or claimed herein or that it alone, or in any combination with any other reference or references, teaches, suggests or discloses any such invention. Further, to the extent that any meaning or definition of a term in this document conflicts with any meaning or definition of the same term in a document incorporated by reference, the meaning or definition assigned to that term in this document shall govern.

While particular embodiments of the present invention have been illustrated and described, it would be obvious to those skilled in the art that various other changes and modifications can be made without departing from the spirit and scope of the invention. It is therefore intended to cover in the appended claims all such changes and modifications that are within the scope of this invention.

What is claimed is:

1. A tampon, comprising:

a pledget, when in an opened configuration, having a first broad surface and second broad surface opposite the first broad surface, each of the first and second broad surfaces being defined by a forward end edge, a rearward end edge, and a pair of opposing longitudinal edges; the first broad surface having a pledget width PW measured between the longitudinal edges; the pledget comprising a mass of fibrous absorbent material disposed within or between one or more layers of an outer cover, the outer cover being formed of a first nonwoven web material;

a withdrawal cord joined to the pledget and trailing from the rearward end edge,

a wicking member comprising a second nonwoven web material that is distinct from the first nonwoven web material and distinct from the withdrawal cord, the wicking member being joined in direct contact with the first broad surface, the wicking member having a wicking member width MW that is less than the pledget width; the wicking member having a trailing portion extending rearwardly from the rearward end of the pledget by a trailing length, and

wherein the withdrawal cord and the wicking member are not coaxial and are not intermingled.

2. The tampon of claim 1 wherein the wicking member comprises at least two plies.

3. The tampon of claim 2 wherein the at least 2 plies both comprise the second nonwoven web material.

4. The tampon of claim 2 wherein the at least 2 plies result from folding the second nonwoven web material along at least one longitudinal fold line.

5. The tampon of claim 1 wherein the wicking member has at least 3 plies.

6. The tampon of claim 5 wherein the second nonwoven web material is folded along longitudinal lines to form at least 3 plies in a roll fold configuration.

7. The tampon of claim 5 wherein the section of second nonwoven web material is folded along longitudinal lines to form at least 4 plies in a "C" fold configuration.

8. The tampon of claim 1 wherein the wicking member is stitched to the pledget.

25

9. The tampon of claim 1 wherein the wicking member and the withdrawal cord are stitched to the pledget.

10. The tampon of claim 9 wherein the wicking member and the withdrawal cord are stitched to the pledget at least in part by common stitches.

11. The tampon of claim 1 wherein the second nonwoven material has a fiber directional bias along a longitudinal direction.

12. The tampon of claim 1 wherein the second nonwoven web material is a spunbond nonwoven web material.

13. The tampon of claim 1 wherein the second nonwoven web material is a carded spunlace nonwoven web material.

14. The tampon of claim 1 wherein the second nonwoven web material comprises fibers selected from the group consisting of polyester fibers, polyethylene fibers, PET fibers, polypropylene fibers, cotton fibers, rayon fibers, viscose fibers, lyocell fibers, and any combination thereof.

15. The tampon of claim 14 wherein the second nonwoven web material comprises no more than about 75 weight

26

percent of fibers selected from the group consisting of cotton fibers, rayon fibers, viscose fibers, lyocell fibers, and any combination thereof.

16. The tampon of claim 14 wherein one or more components of the second nonwoven web material have been treated and thereby rendered hydrophilic.

17. The tampon of claim 1 wherein the second nonwoven web material comprises cotton.

18. The tampon of claim 17 wherein the cotton has been left at least partially unstripped of natural waxes and oils and/or has been treated and thereby reduced in hydrophilicity.

19. The tampon of claim 1 wherein material comprised by the withdrawal cord is hydrophobic.

20. The tampon of claim 1 that exhibits wicking of at least 1.2 grams.

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